PHASE 1 REPORT – Logistics Route Planner

🟦 1. Title

Phase 1: Data Collection and Preprocessing

🟦 2. Objective

The objective of this phase is to collect relevant data for logistics route planning and preprocess it to make it suitable for further analysis and machine learning algorithms.

🟦 3. Introduction

Logistics route planning is an important problem in transportation and supply chain management. It helps in finding the most efficient routes for delivery by minimizing cost, time, and distance.
In this phase, we focus on collecting the dataset and performing preprocessing steps such as cleaning and scaling to prepare the data for further analysis.

🟦 4. Data Collection

The dataset used in this project contains information related to:

- Locations (latitude & longitude)
- Distance between points
- Delivery time
- Customer demand

The dataset is loaded using Python libraries like **pandas**.

🟦 5. Tools & Technologies Used

- Python
- Jupyter Notebook / Google Colab
- Libraries:
    - pandas
    - numpy
    - sklearn

🟦 6. Data Preprocessing

(a) Data Cleaning

- Checked for missing values using isnull()
- Removed or handled missing data using dropna()

(b) Data Transformation

- Converted data into proper format if required

(c) Feature Scaling

- Applied **StandardScaler** to normalize the dataset
- This ensures all features contribute equally

🟦 7. Implementation

The dataset was loaded into a DataFrame and preprocessing steps were applied using Python code.
The cleaned and scaled dataset is now ready for further processing in the next phase.

🟦 8. Output

- Displayed first few rows of dataset
- Verified cleaned data
- Generated scaled dataset

🟦 9. Conclusion

In this phase, we successfully collected and preprocessed the dataset. The data is now clean, structured, and ready for applying machine learning algorithms like clustering in the next phase.

🟦 10. Future Scope

In the next phase, clustering techniques like K-Means will be applied to group similar routes and optimize delivery paths.